Lecture 14: Solving Non-Linear Equations

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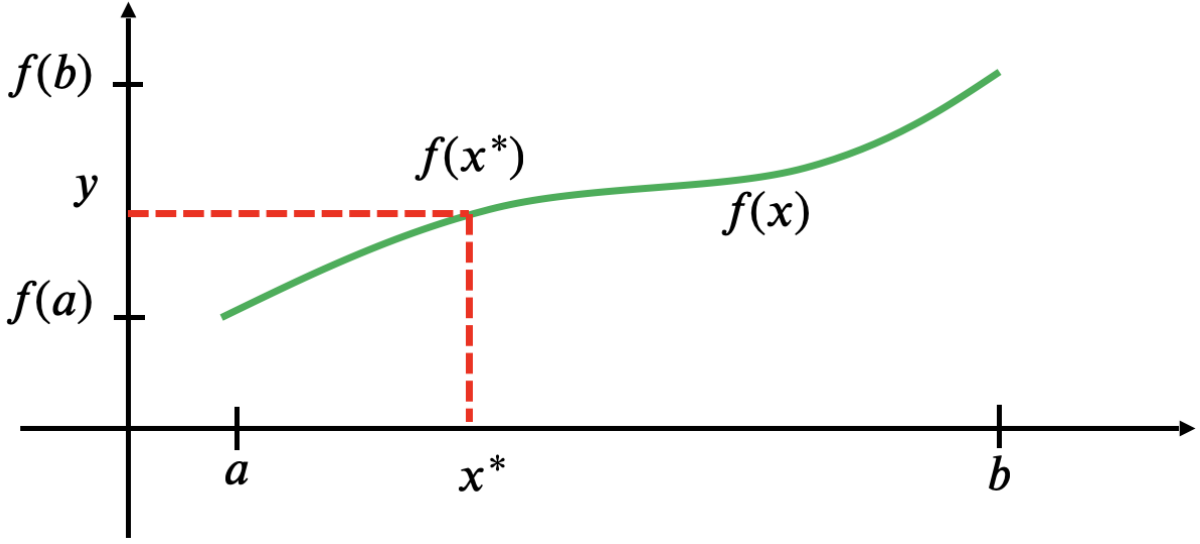
Based on lecture notes by me and many previous CS335 instructors

Fall 2026
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Intermediate Value Theorem

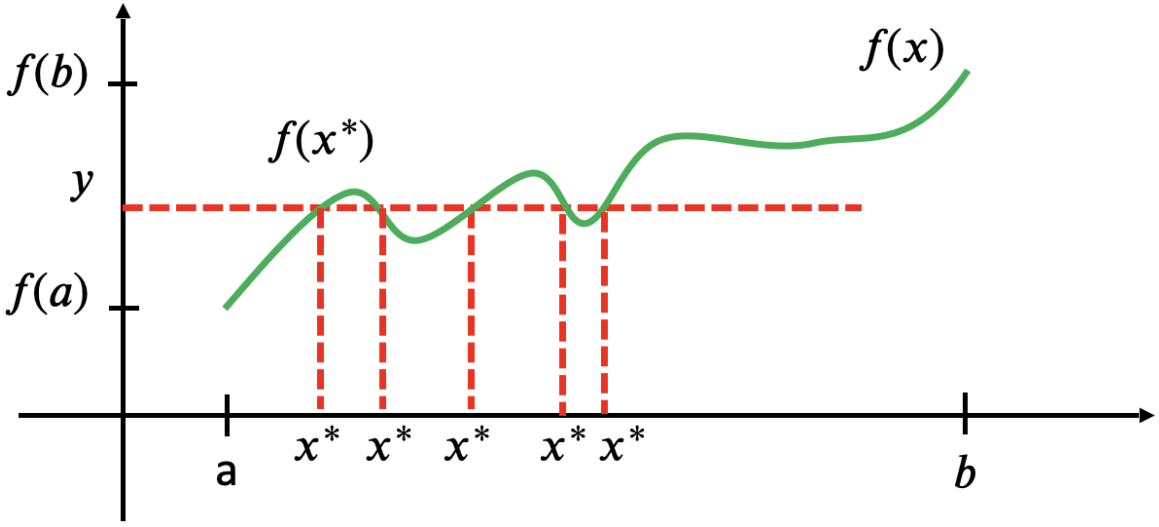
How do we know (in general) where a root may be?

Intermediate Value Theorem: If $f(x)$ is continuous on $[a, b]$ and $y \in [f(a), f(b)]$ then $\exists x^* \in [a, b]$ s.t. $f(x^*) = y$



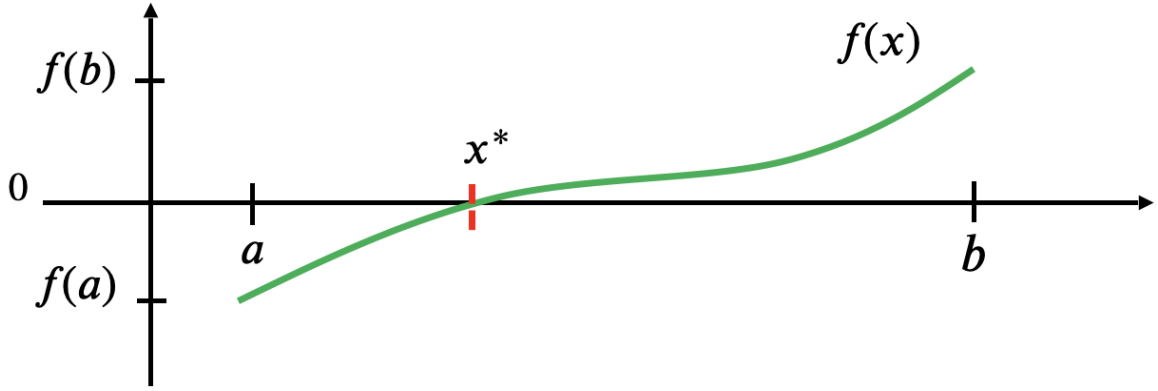
Multiple Roots

There could also be more than one crossing of y in such an interval.



Finding roots using IVT

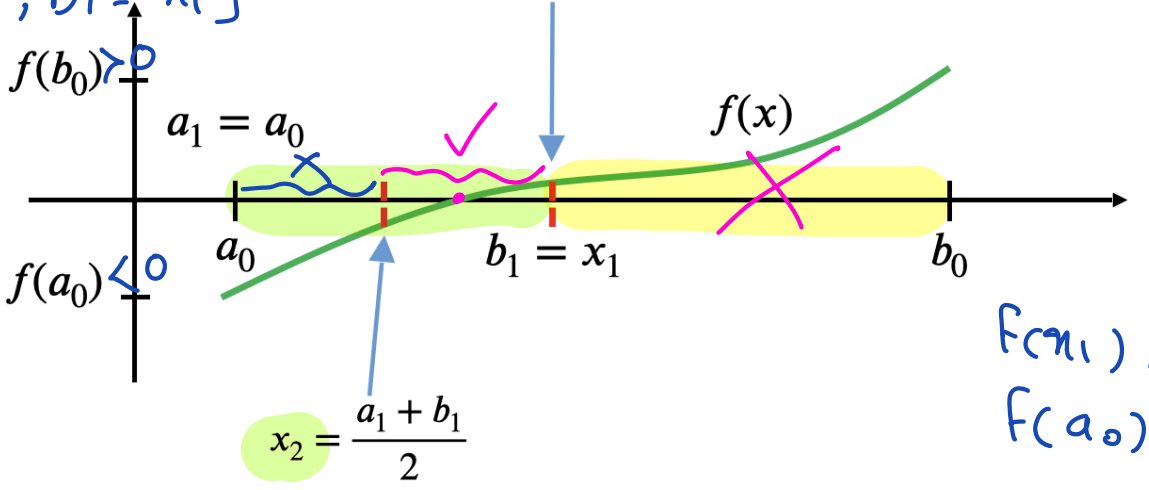
If we can find $[a, b]$ such that the produce $f(a)f(b) < 0$ (i.e., one positive, one negative) then $\exists x^* \in [a, b]$ s.t. $f(x^*) = 0$ since zero is between $f(a)$ and $f(b)$.



This notion leads naturally to our first method, the *bisection method.*

Bisection method

Given $f(x)$ and $[a, b]$, s.t. $f(a)f(b) < 0$, approximate the root with the midpoint $c = \dfrac{a+b}{2}$. If $f(a)$ and $f(c)$ have opposite signs, then repeat (recurse) on the sub-interval $[a, c]$. Otherwise, recurse on $[c, b]$.



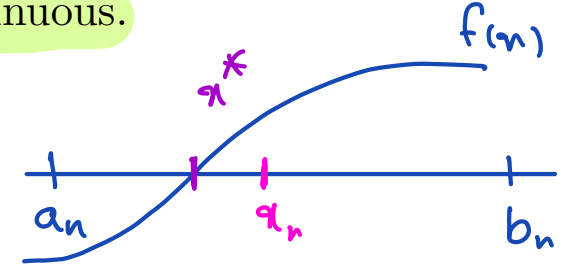
Bisection algorithm

Algorithm Given $[a_0, b_0]$, $f(a_0)f(b_0) < 0$:

1: **for** $k = 0, 1, \ldots$, convergence **do**
2: $\quad x_{k+1} = \dfrac{a_k + b_k}{2}$
3: $\quad$ **if** $f(a_k)f(x_{k+1}) < 0$ **then**
4: $\quad\quad a_{k+1} = a_k$
5: $\quad\quad b_{k+1} = x_{k+1}$
6: $\quad$ **end if**
7: $\quad$ **if** $f(b_k)f(x_{k+1}) < 0$ **then**
8: $\quad\quad a_{k+1} = x_{k+1}$
9: $\quad\quad b_{k+1} = b_k$
10: $\quad$ **end if**
11: **end for**

Notes on Bisection

This method *always* works, provided $f(x)$ is continuous.



The interval length is cut by half at each step.

After n steps the interval becomes $[a_n, b_n]$ where either $a_n = x_n$ or $b_n = x_n$, implying that $|x_n - x^*| < b_n - a_n$.

That is, we always have an error bound for our current solution, x_n. Nice!

Bisection implies that the interval lengths satisfy:

$$|b_n - a_n| = \frac{1}{2^n} |b_0 - a_0|$$

Stopping condition - Bisection

If we stop at $|b_n - a_n| < tol$ for some tolerance value tol, then:

$$2^{-n} |b_0 - a_0| \leq tol$$

$$n \log 2 \geq \log \left(\frac{|b_0 - a_0|}{tol} \right)$$

$$n \geq \frac{1}{\log 2} \left(\log \frac{|b_0 - a_0|}{tol} \right)$$

This gives the minimum number of iterations needed to satisfy the desired tolerance on the error.

Bisection Method Example 1

Question: Show that $f(x) = x^3 + 4x^2 - 10 = 0$ has a root in $[1, 2]$. Use the bisection method to determine an approximation to the root that is accurate to at least within 10^{-1}.

Solution: $f(1) = -5 < 0$, $f(2) = 14 > 0$. Since f is continuous (polynomial), by IVT, there is at least one root in $[1,2]$.

With $a_0 = 1$, $b_0 = 2$, $tol = 0.1$ $\Rightarrow$ $n \geq \log_2\left(\log\left(\frac{|2-1|}{0.1}\right)\right)$

$\Rightarrow$ $n \simeq 3.3219$ $\Rightarrow$ $n = 4$.

k	interval $[a_k, b_k]$	midpoint m_k	$f(m_k)$	next interval
1	$[1, 2]$	1.5	$2.375 > 0$	$[1, 1.5]$
2	$[1, 1.5]$	1.25	$-1.796 < 0$	$[1.25, 1.5]$
3	$[1.25, 1.5]$	1.375	$0.1621 > 0$	$[1.25, 1.375]$
4	$[1.25, 1.375]$	1.3125	$-0.8484 < 0$	$[1.3125, 1.375]$

After 4 iterations, the last interval is $[1.3125, 1.375]$ with width $0.0625 < 0.1$.

$$x^* \approx \frac{1.3125 + 1.375}{2} = 1.34375$$

Bisection Method Example 2

Question: Determine the number of iterations necessary to solve $f(x) = x^3 + 4x^2 - 10 = 0$ with accuracy 10^{-3} using $a = 1, b = 2$.

Solution:

Using the formula $n \geq \frac{1}{\log 2} \cdot \log\left(\frac{|b-a|}{tol}\right)$

with $a = 1$, $b = 2$, $tol = 10^{-3}$, $|b-a| = 1$:

$n \geq 9.9658 \Longrightarrow n = 10$ iteration.

Fixed Point Iteration

Let's consider another method.

If $f(x) = 0$, then clearly $x - x + f(x) = 0$, and thus:

$$x = \underbrace{x - f(x)}_{g(x)}$$

Therefore,

$f(x^*) = 0 \Longrightarrow x^* = g(x^*)$

We call x^* a fixed point of $g(x)$.

This leads to the fixed point iteration algorithm.

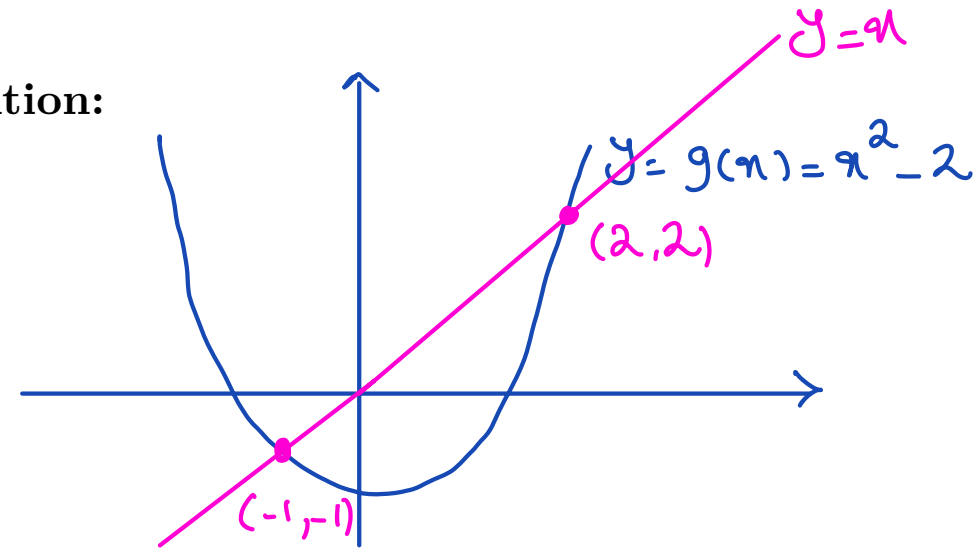
Fixed Point Example

Question: Determine all fixed points of $g(x) = x^2 - 2$.

Solution: A fixed point p of g satisfies

$$p = g(p) = p^2 - 2 \implies p^2 - p - 2 = 0 \implies p = -1, 2$$

Geometric Interpretation:



Fixed Point Iteration Algorithm

Algorithm Given $x_0, g(x)$:

1: **for** $k = 0, 1, \ldots$, convergence **do**
2: $\quad x_{k+1} = g(x_k)$
3: **end for**

$g(x)$ is called an iteration function. Multiple different $g(x)$ are possible for a given $f(x)$, but not all will give a convergent iteration...

Examples

For $f(x) = x^2 - x - 2$, some possibilities might be:

1. $g(x) = x^2 - 2$ ⇝ $x^2 - x - 2 = 0 \Longrightarrow x = x^2 - 2$
   replace x with $g(x)$: $g(x) = x^2 - 2$
2. $g(x) = \sqrt{x+2}$
3. $g(x) = 1 + \frac{2}{x}$
4. $g(x) = x - \frac{x^2 - x - 2}{m}$ for $m \neq 0$.

One can derive such possibilities starting from $f(x) = 0$, and rearranging to isolate an x by itself on the left side.

But which choice of $g(x)$ will work? (More on this later on...)

Newton's method

Yet another method!

Suppose $f(x)$ is continuous with bounded derivatives $f'(x)$, $f''(x)$, and $f'(x) \neq 0$, then (by Taylor series),

$$f(x^*) = f(x_k) + (x^* - x_k) f'(x_k) + O\left((x^* - x_k)^2\right).$$

Ignoring the higher order terms and noting $f(x^*) = 0$,

$$0 \simeq f(x_k) + (x^* - x_k) f'(x_k)$$

$$x^* \simeq x_k - \frac{f(x_k)}{f'(x_k)}$$

Note: This does not give *exact* x^*, because we ignored $O\Big((x^* - x_k)^2\Big)$ terms, so we iterate...

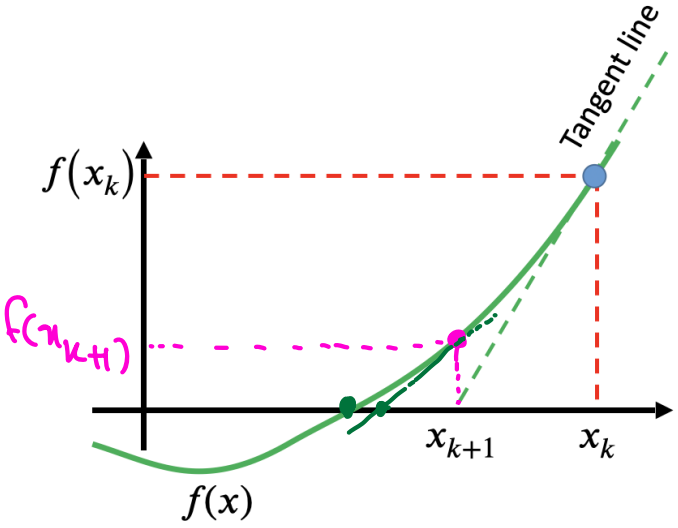
Newton's method: Algorithm

Algorithm Given x_0:

1: **for** $k = 0, 1, \ldots$, convergence **do**

2: $$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

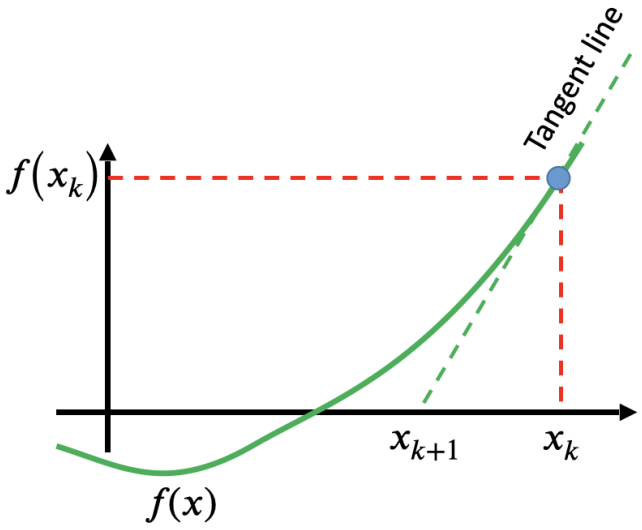
3: **end for**



In essence, we are following the slope at x_k to where it intersects the x-axis, and that becomes the x_{k+1} value.

Newton's method visualization

An equivalent way to understand this is that we are equating the true derivative $f'(x_k)$ with an approximate ("finite difference") derivative, and using that relation to find x_{k+1}.



$$\text{slope} = f'(x_k) \approx \frac{f(x_k) - 0}{x_k - x_{k+1}} \text{ which gives } x_{k+1} = x_k - f(x_k)/f'(x_k)$$

Approximate Derivatives

In practice, $f'(x)$ itself may not be known explicitly (or may be expensive, etc.) For example, if $f(x)$ is a black-box function you can't "look inside".

We can, however, approximate it, which will lead to the **secant method**.

We'll use a finite difference estimate:

$$f'(x_k) \approx \frac{f(x_k + h) - f(x_k)}{h}$$

Then choose $h = x_{k-1} - x_k$, i.e., use *prior step* information:

$$f'(x_k) \approx \frac{f(x_{k-1}) - f(x_k)}{x_{k-1} - x_k}$$

End of Lecture 14

Secant Method

Now replacing the slope $f'(x)$ in Newton's method with this new approximate slope, gives the iteration for the secant method:

$x_{k+1} =$

Notes:

1. We need *two* starting values, x_{k-1}, x_k to get x_{k+1}. Use another method to compute the first step.
2. Secant converges slower than Newton, due to approximating $f'(x_k)$ by finite differences (leading to more error in one step).

Stopping Criteria

For all of these methods, we need to decide when to stop!
Possible options:

1. Maximum number of steps taken - prevent infinite loop
2. Tolerance on the size of the *correction* (change): $|x_{k+1} - x_k| < tol$
3. Tolerance on the side of the function value $|f(x_k)| < tol$

Let's consider using (2) for bisection...

Stopping Criteria (2) on Bisection

From the diagram above: $\left|x_{k+1} - x_k\right| = \dfrac{|b_k - a_k|}{2}$

Stopping criteria $\Rightarrow \dfrac{|b_k - a_k|}{2} < tol \Rightarrow \left|x_{k+1} - x^*\right| < \dfrac{|b_k - a_k|}{2} < tol$

So, for bisection, stopping based on criterion (2) gives a guaranteed error bound.
This is *not* true in general for the Newton, secant, or fixed point methods...
Next time, we'll look at the convergence of these schemes.

Examples - Fixed point and Newton's Methods

Question: Consider the function $f(x) = \cos(x) - x = 0$. Approximate a root of f using:

1. Fixed Point Iteration
2. Newton's Method

Solution:

Example Solution - Fixed Point

Example Solution - Newton's Method

Example Solution - Comparison of both methods